

Task-1

Task-1

On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name Engineering
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name Marketing
Switch(config-vlan)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	Engineering	active	
20	Marketing	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

Task-2

Task-2 Q- Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

```

Switch#
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int rang fast
Switch(config)#int rang fastEthernet 0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan10
Switch(config-if-range)#switchport access vlan10
^
% Invalid input detected at '^' marker.

Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int rang
Switch(config)#int rang fa
Switch(config)#int rang fastEthernet 0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gig0/1, Gig0/2
10   Engineering            active    Fa0/1, Fa0/2
20   Marketing              active    Fa0/3, Fa0/4
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default        active
1005 trnet-default         active
Switch#

```

Task-3

VLAN Configuration

VLAN Configuration Document

VLAN ID: 10

VLAN Name: Engineering

Assigned Ports: FastEthernet0/1 - FastEthernet0/4

VLAN ID: 20

VLAN Name: Marketing

Assigned Ports: FastEthernet0/5 - FastEthernet0/8

Task-4

Two Benefits of VLANs in a College Campus

1. **Better Network Security:** VLANs can separate the Coding Lab and Networking Lab into different logical networks. This prevents students in one lab from
2. directly accessing devices or resources in the other lab.
3. **Reduced Network Traffic:** VLANs divide the campus network into smaller broadcast domains. For example, traffic from the Coding Lab stays mainly within its VLAN, reducing unnecessary traffic and improving network performance.

Conclusion: VLANs provide **better security and improved network performance** by logically separating different college labs.